

Logan Selley

Geospatial Data Engineer | Spatial Analysis & Pipeline Development

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WORK EXPERIENCE

RMSI for Apple

Seattle, Wa

Technical Specialist (SQL)

May 2022 - Present

- Designed multi-stage duplicate cluster detection system using Jaro-Winkler and Levenshtein fuzzy matching for name similarity analysis, spatial clustering for proximity detection, and Union-Find connected component algorithms to identify clusters across vendor datasets
- Co-lead the end-to-end Territories Build pipeline managing team of 10, overseeing territory data lifecycle from vendor handoff through production release
- Developed Python CLI automation suite integrating Snowflake ETL workflows, pre-ingest validation, and data standardization, significantly reducing manual data preparation time and editor-introduced errors
- Led team through dual infrastructure migrations: Postgres to Snowflake for database layer and Hadoop HDFS to AWS S3 for storage layer, rewriting spatial queries and data workflows to leverage cloud-native architectures
- Pioneered complex geometry repair workflows including snap-to-grid alignment, Voronoi tessellation for equitable territory expansion on shared islands, and topology validation using WKB encoding standards
- Performed gap analysis on data quality and internal processes, identifying systematic vendor issues and workflow bottlenecks through pattern recognition in geometry errors, naming inconsistencies, and coverage gaps, drastically reducing downstream issue reports through early validation and systematic data cleanup
- Served as primary technical liaison between vendor teams and Apple stakeholders, translating analytical findings into actionable requirements and quality standards

GIS Technician

June 2021 - May 2022

- Designed and implemented spatial query workflows using buffer analysis, spatial joins, and geometric validation to automate territory boundary alignment and feature relationship detection, eliminating manual review requirements, reclaiming significant analyst time for higher-value work
- Led global country boundary refresh project, developing approach for handling disputed territories through spatial analysis and stakeholder consultation
- Implemented deduplication detection methodology combining distance-based matching, fuzzy string matching (Levenshtein), and spatial overlap analysis to identify duplicate features across regional datasets, automating previously manual workflows

EDUCATION

University of Washington

Seattle Washington

BA Geography: Data Science, minors in Informatics, Music

June 2021

TECHNICAL SKILLS

Programming & Scripting:	Python (Pandas, NumPy, GeoPandas, Shapely, Fiona, Matplotlib, Requests), SQL (Snowflake, PostgreSQL), R (tidyverse, ggplot2, Shiny), JavaScript, Regex
GIS & Spatial Analysis	QGIS, ArcGIS, Leaflet, Mapbox Spatial joins, buffer analysis, topology validation, geometry repair, clustering, Voronoi tessellation, WKB/WKT encoding
Data Quality & Methodology	Fuzzy matching (Levenshtein, Jaro-Winkler), deduplication (Union-Find, connected components), statistical analysis, QA workflow development
Data Visualization	Tableau, R Shiny, Python (Matplotlib), web mapping, geospatial storytelling
Platforms & Tools	Snowflake, PostgreSQL/PostGIS, AWS S3, Hadoop HDFS, Docker, Git, REST APIs